MATH 350-2 Advanced Calculus

W.R. Casper

Department of Mathematics
California State University Fullerton

August 24, 2025

Outline

Outline

Defining the reals

Defining the reals

Question

What are the real numbers?

Defining the reals

Question

What are the real numbers?

- a **set**

Defining the reals

Question

What are the real numbers?

- a **set**
- more specifically: a **field**

Defining the reals

Question

What are the real numbers?

- a **set**
- more specifically: a **field**
- even more specifically: an **ordered field**

Defining the reals

Question

What are the real numbers?

- a **set**
- more specifically: a **field**
- even more specifically: an **ordered field**
- even, even more specifically: a **complete ordered field**

Defining the reals

Question

What are the real numbers?

- a **set**
- more specifically: a **field**
- even more specifically: an **ordered field**
- even, even more specifically: a **complete ordered field**

let's define these things

Field axioms

A **field** is a set F with a way to do addition $+$ and multiplication $\times$ with

Field axioms

A **field** is a set F with a way to do addition $+$ and multiplication $\times$ with

A1 **commutativity**: $x + y = y + x$ and $xy = yx$

Field axioms

A **field** is a set F with a way to do addition $+$ and multiplication $\times$ with

- A1 **commutativity**: $x + y = y + x$ and $xy = yx$
- A2 **associativity**: $(x + y) + z = x + (y + z)$ and $(xy)z = x(yz)$

Field axioms

A **field** is a set F with a way to do addition $+$ and multiplication $\times$ with

- A1 **commutativity**: $x + y = y + x$ and $xy = yx$
- A2 **associativity**: $(x + y) + z = x + (y + z)$ and $(xy)z = x(yz)$
- A3 **distributivity**: $x(y + z) = xy + xz$

Field axioms

A **field** is a set F with a way to do addition $+$ and multiplication $\times$ with

- A1 **commutativity**: $x + y = y + x$ and $xy = yx$
- A2 **associativity**: $(x + y) + z = x + (y + z)$ and $(xy)z = x(yz)$
- A3 **distributivity**: $x(y + z) = xy + xz$
- A4 **additive inverse**: given $x, y \in F$, there exists a unique $z \in F$ with $x = y + z$

$$\text{Notation:} \quad z := x - y$$

Field axioms

A **field** is a set F with a way to do addition $+$ and multiplication $\times$ with

- A1 **commutativity**: $x + y = y + x$ and $xy = yx$
- A2 **associativity**: $(x + y) + z = x + (y + z)$ and $(xy)z = x(yz)$
- A3 **distributivity**: $x(y + z) = xy + xz$
- A4 **additive inverse**: given $x, y \in F$, there exists a unique $z \in F$ with $x = y + z$

$$\text{Notation:} \quad z := x - y$$

- A5 **multiplicative inverse**: given $x, y \in F$ with $y \neq 0$, there exists a unique $z \in F$ with $x = yz$

$$\text{Notation:} \quad z := x/y$$

Field axioms

A **field** is a set F with a way to do addition $+$ and multiplication $\times$ with

- A1 **commutativity**: $x + y = y + x$ and $xy = yx$
- A2 **associativity**: $(x + y) + z = x + (y + z)$ and $(xy)z = x(yz)$
- A3 **distributivity**: $x(y + z) = xy + xz$
- A4 **additive inverse**: given $x, y \in F$, there exists a unique $z \in F$ with $x = y + z$

$$\text{Notation:} \quad z := x - y$$

- A5 **multiplicative inverse**: given $x, y \in F$ with $y \neq 0$, there exists a unique $z \in F$ with $x = yz$

$$\text{Notation:} \quad z := x/y$$

Special notation: $0 := x - x$ and $1 := x/x$, and $-x := 0 - x$

Challenge!

Problem

use the axioms to show that $(x + y)z = xz + yz$.

Challenge!

Problem

use the axioms to show that $(x+y)z = xz + yz$.

Solution

$$(x+y)z \overset{A1}{=} z(x+y) \overset{A3}{=} zx + zy \overset{A1}{=} xz + yz.$$

Challenge!

Problem

use the axioms to show that the value of $x - x$ doesn't depend on x.

Challenge!

Problem

use the axioms to show that the value of $x - x$ doesn't depend on x.

Solution

Suppose $z_1 = x - x$ and $z_2 = y - y$.

Challenge!

Problem

use the axioms to show that the value of $x - x$ doesn't depend on x.

Solution

Suppose $z_1 = x - x$ and $z_2 = y - y$.
Then $x = x + z_1$ and $y = y + z_2$.

Challenge!

Problem

use the axioms to show that the value of $x - x$ doesn't depend on x.

Solution

Suppose $z_1 = x - x$ and $z_2 = y - y$.
Then $x = x + z_1$ and $y = y + z_2$. Therefore

$$(x+y)+z_1 \overset{A2}{=} x+(y+z_1) \overset{A1}{=} x+(z_1+y) \overset{A2}{=} (x+z_1)+y = x+y,$$

Challenge!

Problem

use the axioms to show that the value of $x - x$ doesn't depend on x.

Solution

Suppose $z_1 = x - x$ and $z_2 = y - y$.
Then $x = x + z_1$ and $y = y + z_2$. Therefore

$$(x+y)+z_1 \overset{A2}{=} x+(y+z_1) \overset{A1}{=} x+(z_1+y) \overset{A2}{=} (x+z_1)+y = x+y,$$

$$(x+y)+z_2 \overset{A2}{=} x+(y+z_2) = x+y,$$

Challenge!

Problem

use the axioms to show that the value of $x - x$ doesn't depend on x.

Solution

Suppose $z_1 = x - x$ and $z_2 = y - y$.
Then $x = x + z_1$ and $y = y + z_2$. Therefore

$$(x+y)+z_1 \overset{A2}{=} x+(y+z_1) \overset{A1}{=} x+(z_1+y) \overset{A2}{=} (x+z_1)+y = x+y,$$

$$(x+y)+z_2 \overset{A2}{=} x+(y+z_2) = x+y,$$

so that

$$z_1 = (x+y)-(x+y) = z_2.$$

Challenge!

Problem

use the axioms to show that the value of x/x doesn't depend on x.

Challenge!

Problem

use the axioms to show that the value of x/x doesn't depend on x.

Solution

Suppose $z_1 = x/x$ and $z_2 = y/y$.

Challenge!

Problem

use the axioms to show that the value of x/x doesn't depend on x.

Solution

Suppose $z_1 = x/x$ and $z_2 = y/y$.
Then $x = z_1 x$ and $y = z_2 y$.

Challenge!

Problem

use the axioms to show that the value of x/x doesn't depend on x.

Solution

Suppose $z_1 = x/x$ and $z_2 = y/y$.
Then $x = z_1 x$ and $y = z_2 y$. Therefore

$$z_1(xy) \overset{A2}{=} (z_1 x)y = xy,$$

Challenge!

Problem

use the axioms to show that the value of x/x doesn't depend on x.

Solution

Suppose $z_1 = x/x$ and $z_2 = y/y$.
Then $x = z_1 x$ and $y = z_2 y$. Therefore

$$z_1(xy) \overset{A2}{=} (z_1 x)y = xy,$$

$$z_2(xy) \overset{A2}{=} (z_2 x)y \overset{A1}{=} (x z_2)y \overset{A2}{=} x(z_2 y) = xy,$$

Challenge!

Problem

use the axioms to show that the value of x/x doesn't depend on x.

Solution

Suppose $z_1 = x/x$ and $z_2 = y/y$.
Then $x = z_1 x$ and $y = z_2 y$. Therefore

$$z_1(xy) \overset{A2}{=} (z_1 x)y = xy,$$

$$z_2(xy) \overset{A2}{=} (z_2 x)y \overset{A1}{=} (xz_2)y \overset{A2}{=} x(z_2 y) = xy,$$

so that

$$z_1 = (xy)/(xy) = z_2.$$

Field examples

Lots of examples of fields!

Field examples

Lots of examples of fields!

- **rational numbers**

$$\mathbb{Q} = \{a/b : a, b \text{ integers with } b \neq 0\}.$$

Field examples

Lots of examples of fields!

- **rational numbers**

$$\mathbb{Q} = \{a/b : a, b \text{ integers with } b \neq 0\}.$$

- **real numbers** $\mathbb{R}$ (what are they?)

Field examples

Lots of examples of fields!

- **rational numbers**

$$\mathbb{Q} = \{a/b : a, b \text{ integers with } b \neq 0\}.$$

- **real numbers** $\mathbb{R}$ (what are they?)
- **complex numbers**

$$\mathbb{C} = \{a + ib : a, b \text{ real numbers}\}.$$

Field examples

Lots of examples of fields!

- **rational numbers**

$$\mathbb{Q} = \{a/b : a, b \text{ integers with } b \neq 0\}.$$

- **real numbers** $\mathbb{R}$ (what are they?)
- **complex numbers**

$$\mathbb{C} = \{a + ib : a, b \text{ real numbers}\}.$$

- **field of Boolean numbers**

$$\mathbb{F}_2 = \{0, 1\}$$

$$0+0=0, \quad 0+1=1, \quad 1+0=1, \quad 1+1=0$$

$$0 \cdot 0 = 0, \quad 0 \cdot 1 = 0, \quad 1 \cdot 0 = 0, \quad 1 \cdot 1 = 1$$

Challenge!

Problem

explain why the set of integers isn't a field

Challenge!

Problem

explain why the set of integers isn't a field

Solution

There aren't multiplicative inverses!

Challenge!

Problem

explain why the set of integers isn't a field

Solution

There aren't multiplicative inverses! For example, $1/2$ doesn't make sense because there isn't an integer z with $1 = 2z$.

Order axioms

An **ordered field** is a field F with a relation $<$

Order axioms

An **ordered field** is a field F with a relation $<$

- A6 **trichotomy**: for any $x, y \in F$, exactly one of the following holds:

$$x < y, \quad y < x, \quad \text{or} \quad x = y$$

Order axioms

An **ordered field** is a field F with a relation $<$

- A6 **trichotomy**: for any $x, y \in F$, exactly one of the following holds:

$$x < y, \quad y < x, \quad \text{or} \quad x = y$$

- A7 **compatible with** $+$: $x < y$ implies $x + z < y + z$

Order axioms

An **ordered field** is a field F with a relation $<$

- A6 **trichotomy**: for any $x, y \in F$, exactly one of the following holds:
$$x < y, \quad y < x, \quad \text{or} \quad x = y$$
- A7 **compatible with** $+$: $x < y$ implies $x + z < y + z$
- A8 **compatible with** $\times$: $0 < x$ and $0 < y$ implies $0 < xy$

Order axioms

An **ordered field** is a field F with a relation $<$

- A6 **trichotomy**: for any $x, y \in F$, exactly one of the following holds:

$$x < y, \quad y < x, \quad \text{or} \quad x = y$$

- A7 **compatible with** $+$: $x < y$ implies $x + z < y + z$
- A8 **compatible with** $\times$: $0 < x$ and $0 < y$ implies $0 < xy$
- A9 **transitivity**: $x < y$ and $y < z$ implies $x < z$

Order axioms

An **ordered field** is a field F with a relation $<$

- A6 **trichotomy**: for any $x, y \in F$, exactly one of the following holds:
$$x < y, \quad y < x, \quad \text{or} \quad x = y$$
- A7 **compatible with** $+$: $x < y$ implies $x + z < y + z$
- A8 **compatible with** $\times$: $0 < x$ and $0 < y$ implies $0 < xy$
- A9 **transitivity**: $x < y$ and $y < z$ implies $x < z$

Special notation:

- $x > y$ means $y < x$
- $x \leq y$ means $x < y$ or $x = y$
- $x \geq y$ means $y \leq x$

Challenge!

Problem

explain why $\mathbb{F}_2$ cannot be made into an ordered field

Challenge!

Problem

explain why $\mathbb{F}_2$ cannot be made into an ordered field

Solution

There is no way to put an order relation $<$ on it!

Challenge!

Problem

explain why $\mathbb{F}_2$ cannot be made into an ordered field

Solution

There is no way to put an order relation $<$ on it!
The trichotomy (Axiom 6) would say either $0 < 1$ or $1 < 0$.

Challenge!

Problem

explain why $\mathbb{F}_2$ cannot be made into an ordered field

Solution

There is no way to put an order relation $<$ on it!
The trichotomy (Axiom 6) would say either $0 < 1$ or $1 < 0$.
If $0 < 1$, then by Axiom 7

$$1 = 0 + 1 < 1 + 1 = 0$$

but this means both $0 < 1$ and $1 < 0$, which violates the trichotomy.

Challenge!

Problem

explain why $\mathbb{F}_2$ cannot be made into an ordered field

Solution

There is no way to put an order relation $<$ on it!
The trichotomy (Axiom 6) would say either $0 < 1$ or $1 < 0$.
If $0 < 1$, then by Axiom 7

$$1 = 0 + 1 < 1 + 1 = 0$$

but this means both $0 < 1$ and $1 < 0$, which violates the trichotomy. Same outcome if $1 < 0$.

Apostol Theorem 1.1

Theorem

If $a < b + \epsilon$ for all $\epsilon > 0$, then $a \leq b$.

Apostol Theorem 1.1

Theorem

If $a < b + \epsilon$ for all $\epsilon > 0$, then $a \leq b$.

Proof.

(Contradiction.) Assume the statement is false.

Apostol Theorem 1.1

Theorem

If $a < b + \epsilon$ for all $\epsilon > 0$, then $a \leq b$.

Proof.

(Contradiction.) Assume the statement is false.
Then the trichotomy says $a > b$.

Apostol Theorem 1.1

Theorem

If $a < b + \epsilon$ for all $\epsilon > 0$, then $a \leq b$.

Proof.

(Contradiction.) Assume the statement is false.
Then the trichotomy says $a > b$.
This means $a - b > 0$ by A7.

Apostol Theorem 1.1

Theorem

If $a < b + \epsilon$ for all $\epsilon > 0$, then $a \leq b$.

Proof.

(Contradiction.) Assume the statement is false.
Then the trichotomy says $a > b$.
This means $a - b > 0$ by A7.
This means $(a - b)/2 > 0$ by A8.

Apostol Theorem 1.1

Theorem

If $a < b + \epsilon$ for all $\epsilon > 0$, then $a \leq b$.

Proof.

(Contradiction.) Assume the statement is false.
Then the trichotomy says $a > b$.
This means $a - b > 0$ by A7.
This means $(a - b)/2 > 0$ by A8.
This means

$$a < b + \frac{a-b}{2} = \frac{a+b}{2} \overset{\text{A7\&A8}}{<} \frac{a+a}{2} = a.$$

Apostol Theorem 1.1

Theorem

If $a < b + \epsilon$ for all $\epsilon > 0$, then $a \leq b$.

Proof.

(Contradiction.) Assume the statement is false.
Then the trichotomy says $a > b$.
This means $a - b > 0$ by A7.
This means $(a - b)/2 > 0$ by A8.
This means

$$a < b + \frac{a-b}{2} = \frac{a+b}{2} \overset{\text{A7\&A8}}{<} \frac{a+a}{2} = a.$$

This says $a < a$, which contradicts the trichotomy. $\square$

Outline

Real numbers

The **real numbers** are the unique field $\mathbb{R}$ satisfying Axiom 1-Axiom 9, plus an extra axiom called the completeness axiom.

$$\mathbb{R} = \text{the one and only complete, ordered field}$$

A complete ordered field F satisfies

Real numbers

The **real numbers** are the unique field $\mathbb{R}$ satisfying Axiom 1-Axiom 9, plus an extra axiom called the completeness axiom.

$$\mathbb{R} = \text{the one and only complete, ordered field}$$

A complete ordered field F satisfies

- A10 **completeness axiom**: every set of numbers $S \subseteq F$ which is bounded above has a supremum $\sup(S)$

Real numbers

The **real numbers** are the unique field $\mathbb{R}$ satisfying Axiom 1-Axiom 9, plus an extra axiom called the completeness axiom.

$$\mathbb{R} = \text{the one and only complete, ordered field}$$

A complete ordered field F satisfies

A10 **completeness axiom**: every set of numbers $S \subseteq F$ which is bounded above has a supremum $\sup(S)$

- Understanding what this means is challenging, but will be fundamental to the entire course!

Real numbers

The **real numbers** are the unique field $\mathbb{R}$ satisfying Axiom 1-Axiom 9, plus an extra axiom called the completeness axiom.

$$\mathbb{R} = \text{the one and only complete, ordered field}$$

A complete ordered field F satisfies

- A10 **completeness axiom**: every set of numbers $S \subseteq F$ which is bounded above has a supremum $\sup(S)$
  - Understanding what this means is challenging, but will be fundamental to the entire course!
  - Intuitively, it represents the fact that the real line has no holes or gaps.

Upper bounds

An **upper bound** for a set $S \subseteq \mathbb{R}$ is a number b such that

$$x \leq b \quad \text{for all } x \in S.$$

Upper bounds

An **upper bound** for a set $S \subseteq \mathbb{R}$ is a number b such that

$$x \leq b \quad \text{for all } x \in S.$$

In this case, we say S is **bounded above** by b.

Upper bounds

An **upper bound** for a set $S \subseteq \mathbb{R}$ is a number b such that

$$x \leq b \quad \text{for all } x \in S.$$

In this case, we say S is **bounded above** by b.
If $b \in S$ also, then b is called a **maximal element** of S

Upper bounds

An **upper bound** for a set $S \subseteq \mathbb{R}$ is a number b such that

$$x \leq b \quad \text{for all } x \in S.$$

In this case, we say S is **bounded above** by b.
If $b \in S$ also, then b is called a **maximal element** of S
Examples:

Upper bounds

An **upper bound** for a set $S \subseteq \mathbb{R}$ is a number b such that

$$x \leq b \quad \text{for all } x \in S.$$

In this case, we say S is **bounded above** by b.
If $b \in S$ also, then b is called a **maximal element** of S
Examples:

- 34 is an upper bound of $[-1, 5]$, but not a maximal element

Upper bounds

An **upper bound** for a set $S \subseteq \mathbb{R}$ is a number b such that

$$x \leq b \quad \text{for all } x \in S.$$

In this case, we say S is **bounded above** by b.
If $b \in S$ also, then b is called a **maximal element** of S
Examples:

- 34 is an upper bound of $[-1, 5]$, but not a maximal element
- 5 is a maximal element of $[-1, 5]$

Upper bounds

An **upper bound** for a set $S \subseteq \mathbb{R}$ is a number b such that

$$x \leq b \quad \text{for all } x \in S.$$

In this case, we say S is **bounded above** by b.
If $b \in S$ also, then b is called a **maximal element** of S
Examples:

- 34 is an upper bound of $[-1, 5]$, but not a maximal element
- 5 is a maximal element of $[-1, 5]$
- 3 is an upper bound of $[0, 3)$, but not a maximal element

Upper bounds

An **upper bound** for a set $S \subseteq \mathbb{R}$ is a number b such that

$$x \leq b \quad \text{for all } x \in S.$$

In this case, we say S is **bounded above** by b.
If $b \in S$ also, then b is called a **maximal element** of S
Examples:

- 34 is an upper bound of $[-1, 5]$, but not a maximal element
- 5 is a maximal element of $[-1, 5]$
- 3 is an upper bound of $[0, 3)$, but not a maximal element
- $\mathbb{Z}_+$ has no upper bound

Upper bounds

An **upper bound** for a set $S \subseteq \mathbb{R}$ is a number b such that

$$x \leq b \quad \text{for all } x \in S.$$

In this case, we say S is **bounded above** by b.
If $b \in S$ also, then b is called a **maximal element** of S
Examples:

- 34 is an upper bound of $[-1, 5]$, but not a maximal element
- 5 is a maximal element of $[-1, 5]$
- 3 is an upper bound of $[0, 3)$, but not a maximal element
- $\mathbb{Z}_+$ has no upper bound
- $[3, 7)$ has an upper bound but no maximal element

Challenge!

Problem

Show that if S has a maximal element, then it is unique.

Challenge!

Problem

Show that if S has a maximal element, then it is unique.

Hint: use the definition!

Challenge!

Problem

Show that if S has a maximal element, then it is unique.

Hint: use the definition!

Solution

Suppose that b_1 and b_2 are both maximal elements of S.

Challenge!

Problem

Show that if S has a maximal element, then it is unique.

Hint: use the definition!

Solution

Suppose that b_1 and b_2 are both maximal elements of S.
Then $x \leq b_1$ and $x \leq b_2$ for all $x \in S$.

Challenge!

Problem

Show that if S has a maximal element, then it is unique.

Hint: use the definition!

Solution

Suppose that b_1 and b_2 are both maximal elements of S.
Then $x \leq b_1$ and $x \leq b_2$ for all $x \in S$.
Moreover, $b_1 \in S$ and $b_2 \in S$.

Challenge!

Problem

Show that if S has a maximal element, then it is unique.

Hint: use the definition!

Solution

Suppose that b_1 and b_2 are both maximal elements of S.
Then $x \leq b_1$ and $x \leq b_2$ for all $x \in S$.
Moreover, $b_1 \in S$ and $b_2 \in S$.
Since $b_1 \in S$ and b_2 is an upper bound of S, $b_1 \leq b_2$.

Challenge!

Problem

Show that if S has a maximal element, then it is unique.

Hint: use the definition!

Solution

Suppose that b_1 and b_2 are both maximal elements of S.
Then $x \leq b_1$ and $x \leq b_2$ for all $x \in S$.
Moreover, $b_1 \in S$ and $b_2 \in S$.
Since $b_1 \in S$ and b_2 is an upper bound of S, $b_1 \leq b_2$.
Likewise, since $b_2 \in S$ and b_1 is an upper bound of S, $b_2 \leq b_1$.

Challenge!

Problem

Show that if S has a maximal element, then it is unique.

Hint: use the definition!

Solution

Suppose that b_1 and b_2 are both maximal elements of S.
Then $x \leq b_1$ and $x \leq b_2$ for all $x \in S$.
Moreover, $b_1 \in S$ and $b_2 \in S$.
Since $b_1 \in S$ and b_2 is an upper bound of S, $b_1 \leq b_2$.
Likewise, since $b_2 \in S$ and b_1 is an upper bound of S, $b_2 \leq b_1$.
By the trichotomy, we find $b_1 = b_2$.

Challenge!

Problem

Show that if S has a maximal element, then it is unique.

Hint: use the definition!

Solution

Suppose that b_1 and b_2 are both maximal elements of S.
Then $x \leq b_1$ and $x \leq b_2$ for all $x \in S$.
Moreover, $b_1 \in S$ and $b_2 \in S$.
Since $b_1 \in S$ and b_2 is an upper bound of S, $b_1 \leq b_2$.
Likewise, since $b_2 \in S$ and b_1 is an upper bound of S, $b_2 \leq b_1$.
By the trichotomy, we find $b_1 = b_2$.

Now we can say *the* maximum, $\max(S)$

Supremum

A **supremum** of a set S of real numbers is a real number $b \in \mathbb{R}$ such that

- b is an upper bound of S
- if $b' < b$, then b' is not an upper bound of S

Supremum

A **supremum** of a set S of real numbers is a real number $b \in \mathbb{R}$ such that

- b is an upper bound of S
- if $b' < b$, then b' is not an upper bound of S

In other words

a supremum is a **least upper bound**

Challenge!

Problem

Show that if S has a supremum, then it is unique.

Challenge!

Problem

Show that if S has a supremum, then it is unique.

Hint: use the definition!

Challenge!

Problem

Show that if S has a supremum, then it is unique.

Hint: use the definition!

Solution

Suppose that b_1 and b_2 are both suprema of S.

Challenge!

Problem

Show that if S has a supremum, then it is unique.

Hint: use the definition!

Solution

Suppose that b_1 and b_2 are both suprema of S.
Then b_1 and b_2 are both upper bounds of S.

Challenge!

Problem

Show that if S has a supremum, then it is unique.

Hint: use the definition!

Solution

Suppose that b_1 and b_2 are both suprema of S.
Then b_1 and b_2 are both upper bounds of S.
Since b_1 is a least upper bound, $b_1 \leq b_2$.

Challenge!

Problem

Show that if S has a supremum, then it is unique.

Hint: use the definition!

Solution

Suppose that b_1 and b_2 are both suprema of S.
Then b_1 and b_2 are both upper bounds of S.
Since b_1 is a least upper bound, $b_1 \leq b_2$.
Since b_2 is also a least upper bound, $b_2 \leq b_1$.

Challenge!

Problem

Show that if S has a supremum, then it is unique.

Hint: use the definition!

Solution

Suppose that b_1 and b_2 are both suprema of S.
Then b_1 and b_2 are both upper bounds of S.
Since b_1 is a least upper bound, $b_1 \leq b_2$.
Since b_2 is also a least upper bound, $b_2 \leq b_1$.
By the trichotomy, we find $b_1 = b_2$.

Challenge!

Problem

Show that if S has a supremum, then it is unique.

Hint: use the definition!

Solution

Suppose that b_1 and b_2 are both suprema of S.
Then b_1 and b_2 are both upper bounds of S.
Since b_1 is a least upper bound, $b_1 \leq b_2$.
Since b_2 is also a least upper bound, $b_2 \leq b_1$.
By the trichotomy, we find $b_1 = b_2$.

Now we can say *the* supremum, $\sup(S)$

Challenge!

Problem

Show that if S has a maximum element. Then S has a supremum and $\max(S) = \sup(S)$

Challenge!

Problem

Show that if S has a maximum element. Then S has a supremum and $\max(S) = \sup(S)$

Hint: use the definition!

Challenge!

Problem

Show that if S has a maximum element. Then S has a supremum and $\max(S) = \sup(S)$

Hint: use the definition!

Solution

Let $b = \max(S)$.

Challenge!

Problem

Show that if S has a maximum element. Then S has a supremum and $\max(S) = \sup(S)$

Hint: use the definition!

Solution

Let $b = \max(S)$.
Then b is an upper bound of S and $b \in S$.

Challenge!

Problem

Show that if S has a maximum element. Then S has a supremum and $\max(S) = \sup(S)$

Hint: use the definition!

Solution

Let $b = \max(S)$.
Then b is an upper bound of S and $b \in S$.
If b' is another upper bound of S, then by definition $b \leq b'$.

Challenge!

Problem

Show that if S has a maximum element. Then S has a supremum and $\max(S) = \sup(S)$

Hint: use the definition!

Solution

Let $b = \max(S)$.
Then b is an upper bound of S and $b \in S$.
If b' is another upper bound of S, then by definition $b \leq b'$.
Thus b is the least upper bound of S.

Completeness Axiom

Now we have the machinery in place to state the Completeness Axiom.

Completeness Axiom

Now we have the machinery in place to state the Completeness Axiom.

Completeness Axiom

Now we have the machinery in place to state the Completeness Axiom.

- A10 **completeness axiom**: if S is any subset of real numbers which is bounded above, then it has a supremum $\sup(S)$

Completeness Axiom

Now we have the machinery in place to state the Completeness Axiom.

- A10 **completeness axiom**: if S is any subset of real numbers which is bounded above, then it has a supremum $\sup(S)$

Examples:

Completeness Axiom

Now we have the machinery in place to state the Completeness Axiom.

A10 **completeness axiom**: if S is any subset of real numbers which is bounded above, then it has a supremum $\sup(S)$

Examples:

- 3 is the supremum of $(0, 3)$

Completeness Axiom

Now we have the machinery in place to state the Completeness Axiom.

A10 **completeness axiom**: if S is any subset of real numbers which is bounded above, then it has a supremum $\sup(S)$

Examples:

- 3 is the supremum of $(0, 3)$
- 2 is the supremum of

$$\left\{ \frac{2n}{n+1} : n \in \mathbb{Z}_+ \right\}$$

Completeness Axiom

Now we have the machinery in place to state the Completeness Axiom.

- A10 **completeness axiom**: if S is any subset of real numbers which is bounded above, then it has a supremum $\sup(S)$

Examples:

- 3 is the supremum of $(0, 3)$
- 2 is the supremum of

$$\left\{ \frac{2n}{n+1} : n \in \mathbb{Z}_+ \right\}$$

- 1 is the supremum of

$$\{\sin(n) : n \in \mathbb{Z}_+\}$$